

Q1.

The function $\min(a, b)$ is defined by:

$$\begin{aligned}\min(a, b) &= a, a < b \\ &= b, \text{otherwise}\end{aligned}$$

For example, $\min(7, 2) = 2$ and $\min(-4, 6) = -4$.

Gary claims that the binary operation Δ , which is defined as

$$x \Delta y = \min(x, y - 3)$$

where x and y are real numbers, is associative as finding the smallest number is not affected by the order of operation.

Disprove Gary's claim.

(Total 2 marks)

Q2.

The binary operation $*$ is defined as

$$a * b = a + b + 4 \pmod{6}$$

where $a, b \in \mathbb{Z}$

(a) Show that the set $\{0, 1, 2, 3, 4, 5\}$ forms a group G under $*$;

(5)

(b) Find the proper subgroups of the group G in part (a).

(2)

(c) Determine whether or not the group G in part (a) is isomorphic to the group $K = (\langle 3 \rangle, \times_{14})$

(3)

(Total 10 marks)